

2022-Ph-H1-Q9

Section: Our Dynamic Universe

Topic: Relativity

An astronaut travels at a significant fraction of the speed of light relative to Earth. From the astronaut's perspective, which quantity appears shorter?

Answer: D

(The distance travelled)

Guidance for Students:

This is an example of **length contraction**, a consequence of special relativity. To a moving observer, distances in the direction of motion appear **shorter** than they would to a stationary observer.

Revision Tips:

- Moving clocks run slow: **time dilation**
- Moving rulers contract: **length contraction**
- Formula: $L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$
- Effect only noticeable at high speeds — close to c